and preferred citing his own earlier papers [**47**, p. 310]. He had even less luck with the choice of assistant to write up results on contact transformations and partial differential equations. Felix Hausdorff's interests led him elsewhere, and Lie's thoughts on these subjects were never completely spelled out [**16**, p. 324]. Thus we may never discover the "true Lie".

2.11. Tensor products

In this subsection we recall the notion of tensor product of vector spaces, which will be extensively used below.

Definition 2.11.1. The **tensor product** $V \otimes W$ of vector spaces V and W over a field k is the quotient of the space $V * W$ whose basis is given by formal symbols $v \otimes w$, $v \in V$, $w \in W$, by the subspace spanned by the elements

$$\begin{aligned}
&(v_1 + v_2) \otimes w - v_1 \otimes w - v_2 \otimes w,\\
&v \otimes (w_1 + w_2) - v \otimes w_1 - v \otimes w_2,\\
&av \otimes w - a(v \otimes w),\\
&v \otimes aw - a(v \otimes w),
\end{aligned}$$

where $v \in V, w \in W, a \in k$.

Exercise 2.11.2. Show that $V \otimes W$ can be equivalently defined as the quotient of the free abelian group $V \bullet W$ generated by $v \otimes w$, $v \in V, w \in W$ by the subgroup generated by

$$\begin{aligned}
&(v_1 + v_2) \otimes w - v_1 \otimes w - v_2 \otimes w,\\
&v \otimes (w_1 + w_2) - v \otimes w_1 - v \otimes w_2,\\
&av \otimes w - v \otimes aw,
\end{aligned}$$

where $v \in V, w \in W, a \in k$.

The elements $v \otimes w \in V \otimes W$, for $v \in V, w \in W$ are called pure tensors. Note that in general, there are elements of $V \otimes W$ which are not pure tensors.